DE–2783 **101**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

DIGITAL COMPUTER ORGANIZATION

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

2. (a) Explain how hexodecimal number system is better than other number system. (10)

(b) Discuss about graycode and BCD code. (10)

2. (a) State and prove Demorgan's theorem. (10)

(b) Discuss Quine-Mckluskey method. (10)

2. (a) Describe binary division with example. (10)

(b) Write short notes on :

(i) 2's complement

(ii) 1's complement

(iii) 9's complement

(iv) 10's complement. (10)

2. (a) Discuss demultiplexer with neat diagram. (10)

(b) Explain the function of half subtractor. (10)

2. (a) Write short notes on ALU. (10)

(b) Discuss in detail about full adder. (10)

2. (a) Draw the working of BCD counter. (10)

(b) Explain the working principle of master/slave flip flop. (10)

2. (a) Explain the design of accumulator in detail. (10)

(b) Explain about memory organization in microprocessors. (10)

2. (a) What is meant by instruction cycle and execution cycle? (10)

(b) Explain DMA transfer with the necessary diagram. (10)

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DE–2784 **102**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2001 & 2002 batches)

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

All questions carry equal marks.

1. (a) What is the ethics of information technology? Explain.
   (b) Write a note on communication system.

2. (a) Define database software and explain.
   (b) Discuss about the features of graphics software.

3(a) Define ISDN lines.
   (b) Write down the importance of local networks.

4 Write notes on :
   (a) Cable modems
   (b) E-mail.

5(a) What are the features of DBMS?
   (b) Explain hard disks and optical disks.

6(a) Explain about the different specifiers of scanf ( ) and printf ( ) functions.
   (b) Explain any three decision control statements in C.

7(a) Write short notes on pointers.
   (b) Distinguish between one dimensional arrays and multidimensional arrays.

8(a) Define and explain about recursive functions with an example.
   (b) Write down the advantages of using functions.

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DE–2785 **102**

DISTANCE EDUCATIO

M.C.A. (N-S) DEGREE EXAMINATION,

DECEMBER 2011.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. ($5 \times 20 = 100$)

All questions carry equal marks.

1. (a) What is information? What are its Pre-requisites explain briefly. (5)

(b) What is technology? Write a brief note on the areas where technology is being used. (15)

2. (a) Discuss briefly about the basic features of Interface. (10)

(b) Write a brief note on groupware. (10)

3. (a) Explain the structure of C program. (5)

(b) Write a detailed note on operators. (15)

4(a) Distinguish between structure and union with examples. (10)

(b) Write a C program to perform matrix multiplication using pointers. (10)

5Describe the following

(a) HTML and Active X (10)

(b) Information super highway. (10)

6.(a) Write a C program to calculate the factorial of a given 'N' number using For... loop. (10)

(b) Explain briefly about passing values to a function. (10)

7.(a) Write a C program using pointers, which accepts two strings, compare them and displays which is bigger. (10)

(b) Write short notes on:

(i) Internet (5)

(ii) Internet web browsers. (5)

8. (a) Discuss the important features of XML. (10)

(b) What are the advantages and disadvantages of object oriented programming? Explain briefly. (10)

DE–2786 **103**

DISTANCE EDUCATION

M.C.A. (Non-Semester) DEGREE EXAMINATION, DECEMBER 2011.

C AND DATA STRUCTURE

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the usage of structures in C. (10)

(b) Explain the feature of unions in C. (10)

2. (a) How arrays are used in C? (10)

(b) Explain the concept of array as functions argument. (10)

3.(a) Explain the structure of 'C' program with example. (10)

(b) Explain the formatted I/O operations in details. (10)

4.(a) Write the algorithm for push and pop operations on stack. (10)

(b) Explain the various operations performed on doubly linked list. (10)

5. (a) Explain binary search with an algorithm. (10)

(b) Write the algorithm for addition and deletion operations on queue. (10)

6. (a) Explain various characters I/O functions available with example. (10)

(b) Explain the evaluation of postfix expansion using stack and evaluate the following (10)

(i) 10 5 + 30 20 –1

(ii) 20 10 5 50 – +

7. (a) List out some of the sorting techniques. Explain how bubble sort is applied to sort a given list. (10)

(b) Explain about trees and their applications. (10)

8. (a) Explain the features of pointers and arrays with example. (10)

(b) Explain the functions of loop control structures with example. (10)

DE–2787 **104**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

OFFICE AUTOMATION

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) What are the various components (title bar, menu bar...) of windows? Explain. (10)

(b) What do you mean by folder? How to manipulate it? Explain. (10)

2. (a) How to open, close and save a document? Explain in detail. (10)

(b) What is Icon? How to create it? Explain. (10)

3. (a) How to format a word document? Explain.

(b) What is table? How to create and format a table? Explain.

4. (a) What are the various methods to insert a picture with word document? Explain.

(b) Explain various features of format menu.

5. (a) How to prepare worksheet? Explain its functions with neat diagram.

(b) (i) How to insert and delete rows and columns in Excel?

(ii) How to change a column width? Explain.

6. (a) Explain about chart wizard.

(b) Give the various steps to access form creation.

7. (a) What is query? Explain various methods to create a query.

(b) How to create table with primary key in Access?

8. (a) Give the various functions used to prepare enhanced Power Point Presentation.

(b) How to merge Access table with WORD letter? Explain.

DE–2788 **105**

DISTANCE EDUCATION

M.C.A. (N.S) DEGREE EXAMINATION, DECEMBER 2011.

DISCRETE MATHEMATICS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Show that $\neg(P \wedge Q) \to (\neg P \vee (\neg P \vee Q)) \Leftrightarrow (\neg P \vee Q)$. (10)

   (b) Obtain the principal disjunctive normal form of $P \to ((P \to Q) \wedge \neg(\neg Q \vee \neg P))$. (10)

2. (a) Show that SVR is tautologically implied by $(P \vee Q) \wedge (P \to R) \wedge (Q \to S)$. (10)

   (b) Show that $(x)(P(x) \vee Q(x)) \Rightarrow (x)P(x) \vee (\exists x)Q(x)$. (10)

3. (a) Show by means of an example that $A \times B \neq B \times A$ and $(A \times B) \times C \neq A \times (B \times C)$. (10)

   (b) If $M_k = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$, $M_s = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$ are the rational matrices, find $M_{R \circ S}, M_{\tilde{R}}, M_{\tilde{S}}, M_{R \tilde{\circ} S}$ and show that $M_{R \tilde{\circ} S} = M_{\tilde{S} \circ \tilde{R}}$. (10)

4. (a) If $f(x) = x + 2, g(x) = x - 2$ and $h(x) = 3x$ for $x \in R$, where R is the set of real numbers. Find $g \circ f$; $f \circ g$; $f \circ f$; $g \circ g$; $f \circ h$; $h \circ g$; $h \circ f$ and $f \circ h \circ g$. (10)

   (b) Show that $f : X \to Y$ is one-to-one iff any proper subsets of X are mapped into proper subsets of Y; that is if $A \subset B \subseteq X$, then $f(A) \subset f(B) \subseteq Y$. (10)

5. (a) Given the algebraic system $\langle N, + \rangle$ and $\langle Z_4, +4 \rangle$, where N is the set of natural numbers and + is the operation of addition on N, Show that there exists a homomorphism from $\langle N, + \rangle$ to $\langle Z_4, +4 \rangle$. (10)

   (b) For any commutative monoid $\langle M, * \rangle$, prove that the set of idempotent elements of M, forms a submonoid. (5)

   (c) Show that the set of all invertible elements of a monoid form a group under the same operation as that of the monoid. (5)

6. (a) Show that in a group $\langle G, * \rangle$, if for any $a, b \in G$, $(a * b)^2 = a^2 * b^2$, then $\langle G, * \rangle$ must be abelian. (6)

   (b) Show that the set of all elements a of a group $\langle G, * \rangle$ such that $a * x = x * a$ for every $x \in G$ is a subgroup of G. (7)

   (c) Prove that, the order of a subgroup of a finite group divides the order of the group. (7)

7. (a) State and prove the fundamental theorem of group homomorphism. (9)

   (b) Define field. (3)

(c) Prove that the ring homomorphism preserves the distributive property. (8)

8. (a) Define the length of the path. (3)

(b) Show by means of an example that a simple diagraph in which exactly one node has indegree '0' and every other mode has indegree '1' is not necessarily a directed tree. (10)

(c) In a simple digraph $G, = \langle V, E \rangle$, prove that, every node of the digraph lies in exactly one strong component. (7)

DE–2789 **108**

DISTANCE EDUCATIO

M.C.A. (NS) DEGREE EXAMINATION,

DECEMBER 2011.

OBJECT ORIENTED PROGRAMMING AND C++

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. ($5 \times 20 = 100$)

All questions carry equal marks.

1. (a) Discuss object-oriented programming paradigm.

(b) What are the benefits of OOP?

2. (a) Explain any three control statement used in C++. Give examples.

(b) Illustrate with examples the standard I/O statement.

3. (a) What are the technique features of C++ function?

(b) List the operators in C++. Explain any four operators with example.

4. (a) What are constructors? Illustrate multiple constructors in a class.

(b) How to define member function? Illustrate with an example.

5. (a) What are the advantages and problems with reference feature of pointers?

(b) Write a C++ program to average a list of numbers in ascending order using pointers.

6. What is meant by overloading? What are the rules for overloading operators? Write an example program for operator overloading.

7. (a) Discuss about type conversion.

(b) Write a C++ program to demonstrate type conversion of one class to another class.

8. Write notes on :

(a) Reusing code with member objects. (8)

(b) Storing objects on disk (6)

(c) A list that can save and retrieve itself. (6)

DE–2790 **109**

DISTANCE EDUCATIO

M.C.A. (NS) DEGREE EXAMINATION,

DECEMBER 2011.

COMPUTER GRAPHICS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. ($5 \times 20 = 100$)

All questions carry equal marks.

1. (a) Explain the working of video display unit. (10)

(b) Explain briefly about the various types of input devices. (10)

2. (a) Discuss the simple DDA algorithm. (10)

(b) Explain in detail about Bresenham's algorithm. (10)

3. (a) Explain in detail about the basic transformation about 2D. (10)

(b) Discuss in detail about convex polygon clipping. (10)

4. (a) Discuss about the composite transformation in two dimensional transformations. (10)

(b) Write short notes on

(i) Point Clipping. (5)

(ii) Line Segment Clipping. (5)

5. (a) Explain the basic 3D transformations. (10)

(b) Discuss in detail about the three dimensional rotations. (10)

6. (a) Explain the Matrix and Composite transformation in 3D. (10)

(b) Describe the image space method of hidden surface algorithm. (10)

7. (a) Define Clipping. Discuss Cohen – Sutherland algorithm for the clipping. (10)

(b) Explain the clipping against a finite view volume in 3D. (10)

8. (a) Explain the Styles of command languages. (10)

(b) Explain in detail about the information display in interface design. (10)

DE–2791 **110**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

DESIGN AND ANALYSIS OF ALGORITHM

(2001 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. Write short notes on :

(a) Problem statement (10)

(b) Design of an algorithm. (10)

2. (a) What is structured programming? Explain. (10)

(b) Explain knight's tour in the game of chess. (10)

3. (a) Explain about linked list. Write an algorithm for performing operations on a linked list. (10)

(b) Make a note on adjacency lists. (10)

4. (a) Explain the concept of Heuristics. (10)

(b) Explain the greedy travelling salesman algorithm. (10)

5. (a) Explain backtracking with bicycle lock problem. (10)

(b) Make a note on hill climbing. (10)

6. (a) Make a note on recursion. (10)

(b) Explain Ackerman's function. (10)

7. (a) Make a note on stack and queue. (10)

(b) Explain BFS algorithm in detail. (10)

8. (a) Write a note on sorting. (10)

(b) Explain searching. (10)

DE–2792 **111**

DISTANCE EDUCATION

M.C.A. (N.S) DEGREE EXAMINATION, DECEMBER 2011.

ACCOUNTING AND FINANCIAL MANAGEMENT

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. (5 × 20 = 100)

1. (a) Explain different kinds of cash book.

(b) What the differences between books of original entry and ledger?

2. Define accounting distinguish between book keeping and accounting.

3. From the following transactions extracted from the books of accounts of a manufacturing concern as on 31st December 2004, work out

(a) Consumption values of raw material in the month and

(b) Value of closing stock as on 31st December 2004, under the following two methods of pricing issues.

(i) FIFO

(ii) LIFO.

Show the results in a tabulated from

2004			Quantity in Units	Rate Per unit in rupees
December	1	Opening stock	300	9.70
	3	Purchase	250	
	11	Issue	400	9.80
	15	Purchase	300	
	20	Issues	210	10.05
	25	Purchases	150	
	29	Issues	100	10.30

4. From the following puji muresh confectionery limited the profit of 1998 and 1999 are given below together with expenses :

	1998 (Rs.)	1999 (Rs.)
Materials consumed	1,00,000	1,40,000
Wages	80,000	1,20,000
Overheads		
Fixed	30,000	32,000
Variable	24,000	34,000
Net profit	10,000	20,500

The wage rate was increased by 20% in 1999 than in 1998, similarly materials price were high by 10% in 1999. Analyse the causes of increases in profit.

5. "LMN" Limited Sets its product at Rs. 3 per unit. The company uses a First-in First-out actual costing system. A new fixed manufacturing over head allocation rate is

computed each year by dividing the actual fixed manufacturing overhead cost by the actual production costs.

The following simplified data are related to its first two years of operation :

Units data	Year I	Year II
Sales	1,000	1,200
Production cost	1,400	1,000
Cost		
Variable manufacturing	700	500
Fixed manufacturing	700	700
Variable marketing and administration	1,000	1,200
Fixed Marketing and administration	400	400

Required :

(a) Prepare income statements based on

(i) Absorption costing and

(ii) Variable costing for each year.

(b) Give reasons for the differences in the answer.

6. A concern manufacturing product A, has furnished you with the following information :

	Rs.
Sales	75,000
Direct Materials	30,000
Variable overheads	10,000
Direct labour	10,000
Fixed overhead	15,000

In order to increase its sales by Rs. 25,000 the concern proposes to introduce the product B and estimates the costs in conection there with as follows :

	Rs.
Direct materials	10,000
Direct labour	8,000
Variable overhead	5,000
Fixed overhead	NIL

Advise whether the product B will be profitable or not.

7. Transnational Tractors Ltd. have an installed capacity of 5,000 tractor per annum. for the coming year they have budgeted as follows :

Production/sales 4000 units

Costs	Rs. in crores

(a)	Direct materials	8.00
(b)	Direct wages	0.60
(c)	Factory expenses	0.80
(d)	Administrative expenses	0.20
(e)	Selling expenses	0.20
	Profit	1.00

Factory expenses as well as selling expenses are variable to the extent of 20% calculate the Break Even Capacity utillisation percentage.

8. ABC Ltd. Prepared a budget for the production of a Lakh units of the only commodity manufactured by them for a costing period as under :

	Rs. (Lakh)
Raw materials	2.52
Direct labour	0.75
Direct expenses	0.10
Works over heads (60% fixed)	2.50
Administrative overheads (80% fixed)	0.40
Selling overheads (50% fixed)	0.20

The actual production during the period was only 60,000 units.

Calculate the revised budgeted cost per unit.

DE–2793 **112**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

COMMUNICATION SKILLS

(2001 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Bring out the importance and the process of communication.
2. Write an essay on the importance of developing listening skills.
3. Explain the types of interviews and their intricacies.
4. Discuss the purpose and process of group discussion.
5. Describe body language as a powerful communication.
6. How do you plan for a presentation? Discuss in detail.
7. Negotiations are mandatory in business – Justify.
8. Write short notes on the following :
   (a) Mock interview
   (b) Telephonic conversation
   (c) Communication skills for meetings
   (d) Linear reading.

DE–2794 **201**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

RESOURCE MANAGEMENT TECHNIQUES

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Solve by simplex method

Maximize $Z = 6x_1 + 8x_2$

Subject to

$$5x_1 + 2x_2 \le 20$$
$$x_1 + 2x_2 \le 20$$
$$x_1, x_2 \ge 0.$$

(b) What are the phases involved to optimize resource technique? Explain.

2. (a) Solve the following transportation problem using least cost method.

10	20	5	7
13	9	12	8
4	15	7	9
14	7	1	0
3	12	5	19

(b) Write a step by step procedure of simplex method.

3. (a) Discuss in detail about the Markovian queues.

(b) Indicate some of the queuing models where you would use simulation.

4. (a) Explain briefly about uniform, normal and binomial random variates.

(b) Discuss about the quantity discounts in Inventory models.

5. Explain in detail about the various types of inventory models.

6. (a) Discuss about the individual replacement and group replacement policies.

(b) Write short notes on :

(i) Time chart

(ii) Project control

(iii) Reliability.

7. (a) Draw a network diagram for the below activity.

Activity	Time
1–2	2
1–3	4
1–4	3
2–5	1
3–5	6
4–6	5
5–6	7

(b) Write short notes on :

(i) Node

(ii) Float

(iii) Arrow.

8. (a) Discuss about the variable maintenance costs and fixed money value.

(b) Explain briefly about the cost considerations in project scheduling.

DE – 2795 **202**

DISTANCE EDUCATION

M.C.A.(NS) DEGREE EXAMINATION,

DECEMBER 2011

OPERATING SYSTEMS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Define operating system. Explain its history. (10)

(b) Discuss the structure of operating system. (10)

2. (a) What is semaphore? Explain the operations of semaphore mechanism using binary and counting semaphore. (10)

(b) Write short notes on:

(i) Race condition (5)

(ii) Message passing. (5)

3. What is processor scheduling? What are the various types of scheduling? Explain in detail. (20)

4. (a) Write a note on:

(i) Device controllers (5)

(ii) Device drivers. (5)

(b) Explain how deadlock can be prevented and avoided. (10)

5. (a) Discuss about Device–Independent I/O software. (8)

(b) Explain how memory can be managed without swapping or paging. (12)

6. (a) Explain multiprogramming with fixed and variable partitions. (10)

(b) Explain any two page replacement algorithms. (10)

7. (a) Discuss the memory management with Bit-maps and Buddy system. (10)

(b) Explain various file operations. (10)

8. (a) Write a note on:

(i) Directories (5)

(ii) Shared files. (5)

(b) Explain file protection mechanisms. (10)

DE–2796 **203**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

RDBMS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Compare the database management system with a file system. (10)

   (b) Write detailed notes on user interfaces and explain the process involved in designing the user interface. (10)

2. (a) What are the operations performed on database management system? Explain in detail. (10)

   (b) What is meant by relation? Explain the relational model in detail. (10)

3. (a) Write short notes on the following :
   (i) entity
   (ii) attribute
   (iii) relationship
   (iv) key
   (v) normalisation. (10)

   (b) Write notes on different types of databases and explain the properties of relational database. (10)

4. (a) Write detailed notes on SQL * PLUS environment. (10)

   (b) What are the data types available in oracle? Explain them with suitable examples. (10)

5. (a) What is constraint? Explain different constraints with suitable examples. (10)

   (b) Explain detailed notes on oracle functions. (10)

6. (a) How the data manipulation is performed in oracle database? Explain in detail. (10)

   (b) What do you mean by join? What are the different joins are available in oracle? Explain in detail. (10)

7. (a) Explain the following :
   (i) Privilege
   (ii) Data control language
   (iii) View
   (iv) Precompiler Union. (10)

   (b) Write detailed notes on subqueries? Give suitable examples. (10)

8. (a) What is the use of embedded SQL? Explain in detail. (10)

   (b) Why do we need objects? What are the objects available in oracle? Explain in detail. (10)

DE–2797 **204**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

UNIX AND SHELL PROGRAMMING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20= 100)

1 Explain any five common file system commands in Unix. (6)

(b) Explain grep command with an example. (6)

(c) Discuss the features of stream editor sed. (8)

2) Explain any two shell variables and its use with examples. (10)

(b) Explain the format of for statement and explain its use with an example shell program. (10)

(3) What is Inodes in Unix? Explain. (6)

(b) Explain the method of handling devices in Unix. (6)

(c) Discuss the Unix directory structure. (8)

(4) Explain the while loop with an example program. (10)

(b) Discuss the use of trap command with an example. (10)

(5) Explain the commands zap, pick and news. (10)

(b) Discuss the use of getchar ( ) and putchar ( ) with an example program. (10)

(6) Explain the method of executing the program within another. (10)

(b) Explain the system call signal with an example. (10)

7 Explain the following with respect to implementation of four function calculator

(a) Backus – Naur Form

(b) Parser

(c) Lexical analyzer

(d) Stack machine. (4 × 5 = 20)

(8) Explain any ten MS macro package formatting commands. (10)

(b) Discuss the method of producing tables and equations using tbl and eqn preprocessors. (10)

DE–2798 **205**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

COMPUTER NETWORKS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal Marks.

$(5 \times 20 = 100)$

1. (a) Define Network. Write the importance of computer Network. (10)

(b) Discuss the different transmission media used for data communication. (10)

2. (a) Write short notes on Network architecture. (10)

(b) Explain the features of ISDN. (10)

3. (a) Explain carrier sense multiple Access protocol. (10)

(b) Describe protocol specification and verification. (10)

4. (a) Define CRC. Explain CRC generator and CRC checks with an example. (10)

(b) Write short notes on collision Free Protocols. (10)

5. (a) State the difference between vector Routing and Link State Routing. (10)

(b) Explain the design issues of Network Layers. (10)

6. (a) Discuss about the IP format and addressing schemes. (10)

(b) Write short notes on :

(i) Tunneling. (5)

(ii) Multicasting. (5)

7. (a) Discuss about the error control and flow control. (10)

(b) Explain the elements of Transport protocol. (10)

8. (a) Describe in detail about simple network management protocol. (10)

(b) Explain the basic aspects of electronic mail system. (10)

DE–2799 **208**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

SIMULATION AND MODELLING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write short notes about continuous and discrete system. (10)

   (b) Explain the dynamic physical models. (10)

2. (a) What are the various types of system simulation? Explain. (10)

   (b) Explain the simulation of an Autopilot. (10)

3. (a) Discuss about the modified exponential growth model. (10)

   (b) Describe about the system dynamics diagrams. (5)

   (c) List the different types of system simulation. (5)

4. (a) Discuss in detail about the measures of probability function. (10)

   (b) Write short notes on mathematical solutions of queuing problems. (10)

5. (a) Describe the Dynamo language. (10)

   (b) Write short note on logistic curves. (10)

6. (a) Define SIMSCRIPT. (5)

   (b) Explain about the arrival event, the disconnect event and the closing event of SUBSCRIPT. (15)

7. (a) Explain the GPSS programs with its general description. (10)

   (b) Discuss in detail about the simulation of a manufacturing shop. (10)

8. (a) Explain the delayed calls. (10)

   (b) Define the term transit times. (5)

   (c) Describe the trace drive simulation. (5)

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DE–2800 **209**

DISTANCE EDUCATI

M.C.A. DEGREE EXAMINATION, NOVEMBER 2011.

OBJECT ORIENTED ANALYSIS AND DESIGN

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. (5 × 20 = 100)

All questions carry equal marks.

1. (a) What is an object? Explain its nature. (10)

   (b) Describe various kinds of relationships among classes. (10)

2. (a) Explain the roll of a class in object oriented system development? (10

   (b) What are the criteria to be considered in designing quality classes? Explain. (10)

3. (a) What is UML? Explain the objectives of modeling. (10)

   (b) Explain various steps involved in the view process. (10)

4. Write short notes on

   (a) Association relationship (10)

   (b) Deployment diagram. (10)

5. Draw the class diagram, use-case diagram, interaction diagram for Library management system. (10)

6. (a) Explain the various object oriented analysis approaches. (10)

   (b) Write short notes on key abstraction mechanism. (10)

7. (a) What are the four kinds of abstractions? Explain. (10)

   (b) How to identify the proper classes and objects-Explain. (10)

8. (a) What is meant by use cases'? Explain the use case relationships with example. (10)

   (b) Explain the importance and features of Interaction diagrams. (10)

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DE–2801 **210**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

INTERNET PROGRAMMING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the TCP/IP Protocol Model. (10)

   (b) Discuss about the public and private key encryption. (10)

2. (a) What are the benefits and negatives of using CGI? (5)

   (b) What are benefits of Java? Explain. (5)

   (c) Write a note on implementation of Microsoft Internet explorer. (10)

3. (a) Discuss in detail about the I.P. address and domain names. (6)

   (b) Give brief details about using perl in Internet applications. (8)

   (c) Give the applications of the HTML. (6)

4. (a) Explain the various conversion function in VB script. (10)

   (b) What is Shock wave? Explain. (10)

5. (a) Explain the structure of SGML. (10)

   (b) Explain any six tags in the HTML. (10)

6. (a) Discuss the creation of a Custom Integrated Application with multiple protocols. (10)

   (b) Write short note on Netscape Navigator Plug-Ins. (10)

7. (a) List the benefits of using ActiveX Control (10)

   (b) Discuss the tables and frames in HTML. (10)

8. (a) How to embed audio and video applications into a HTML document? Explain. (10)

   (b) Design a simple web page for a various courses in the Department of computer science using HTML. (10)

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DE– 2802 **211**

DISTANCE EDUCATION

M.C.A (N.S) DEGREE EXAMINATION, DECEMBER 2011.

SOFTWARE ENGINEERING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Explain Evolutionary software process model.
2. Discuss the Risk management for software project.
3. Explain the software configuration management and software reviews.
4. Describe the system requirements analysis.
5. Explain the different types of Design methods.
6. Explain the following :

(a) Fourth Generation Technique.

(b) Decomposition Techniques.

7. Discuss the following :

(a) Software prototyping

(b) White Box Testing and Basic path Testing

8. Explain the various software testing methods with examples.

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DE–2803 **212**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

VISUAL PROGRAMMING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write a note on software development kit (SDK) tools. (10)

   (b) Explain dynamic linking libraries. (10)

2. (a) Explain looping statement in VB with an example. (10)

   (b) What are the types of procedures used in VB. Explain with an example. (10)

3. (a) Differentiate between implicit declaration and explicit declaration with an example. (10)

   (b) Explain briefly about event procedure in VB. (10)

4. (a) Explain INPUT and OUTPUT functions in VB with an example. (10)

   (b) How you add controls to form? Explain with an example. (10)

5. (a) How you write a code for menu controls? Explain. (8)

   (b) Explain the steps to create a database using visual data manager. (12)

6. (a) How dialog boxes used in VC++? Explain. (10)

   (b) Explain how you import VBX controls. (10)

7. (a) Write a note on splitter windows. (10)

   (b) Explain control statements in VC++ with an example. (10)

8. (a) Explain about object linking and embedding (OLE). (10)

   (b) Explain the concept of debugging. (10)

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DE–2804 **301**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

SOFTWARE PROJECT MANAGEMENT

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Describe the activities and their relationships in software model. (10)

(b) Explain how the process can be improved in the software development. (10)

2. (a) Write short notes on :

(i) Personality traits

(ii) Documentation. (10)

(b) Describe the bottomup planning process. (10)

3. (a) Give the importance of vendor relationship in marketing issues. (10)

(b) What is the need for measuring the performance at regular intervals? Explain. (10)

4. (a) Discuss briefly about recovery plans. (10)

(b) Explain the unit test with its objectives and performance. (10)

5. Discuss in detail about project tracking. (20)

6. (a) How can you obtain the product requirements? Explain. (10)

(b) Write a brief note on product objectives. (10)

7. (a) Briefly discuss about function test and function test plan. (10)

(b) How the qualities weak link can be anticipated? Describe. (10)

8. (a) Describe the vendor contract process. (10)

(b) Write down the legal considerations in the vendor contract process. (10)

DE–2805 **302**

DISTANCE EDUCATION

M.C.A. (N.S) DEGREE EXAMINATION, DECEMBER 2011.

CLIENT SERVER TECHNOLOGY

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the evolution of client/server computing. (5)

   (b) Write short notes on downsizing. (5)

   (c) Discuss in detail about client/server development tools. (10)

2. How to prepare a client/server network in a unix operating system? Discuss on its operation and its architecture. (20)

3. Describe the following :

   (a) Client/server systems development methodology. (10)

   (b) Project management. (10)

4. Explain in detail on CORBA architecture and its components. (20)

5. Write short notes on the following :

   (a) Work bench Architecture. (5)

   (b) SQL windows. (10)

   (c) Power builder. (5)

6. Describe the server functionality in detail. (20)

7. Write a detailed note on windows NT server operating system. (20)

8. Sketch the architecture of DCOM and explain the functions of its components with neat sketch. (20)

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DE–2806 **303**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

PATTERN RECOGNITION AND IMAGE PROCESSING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain Real time capture of an image. (8)

(b) Discuss about Ranging devices and calibration. (12)

2. (a) Write about three dimensional imaging. (10)

(b) Describe about Multi image operation. (10)

3. (a) Write in detail about Two dimensional geometric transformation. (10)

(b) What is Back edge detection? Explain pyramid edge detection. (10)

4. (a) Give the need for image compression. (10)

(b) Discuss on Image standards. (10)

5. (a) Write in detail about contour coding. (10)

(b) Explain basic concepts of pattern Recognition. (10)

6. (a) Discuss on automatic pattern recognition system. (10)

(b) Describe Linear decision functions in detail. (10)

7. (a) What is pattern classification? Explain minimum distance pattern classification. (10)

(b) Explain single and multi prototypes in detail. (10)

8. (a) Discuss on duster seeking algorithm. (10)

(b) Explain concepts of Formal language theory. (10)

DE–2807 **304**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION,
DECEMBER 2011.
COMPILER DESIGN
(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Discuss in detail about the structure of a compiler. (20)
2. (a) Explain in detail about lexical analyzer. (10)
   (b) Describe about parse tree with an example. (10)
3. Discuss in detail about the Top–down parsing with example. (20)
4. (a) Explain ambiguous grammar with an example. (10)
   (b) Discuss in detail about LR parsers. (10)
5. (a) Explain in detail about postfix Translations. (15)
   (b) Discuss about intermediate code. (5)
6. (a) Discuss in detail about symbol table and its contents. (10)
   (b) Describe about semantic Errors. (10)
7. Explain in detail about basic blocks represented by DAG. (20)
8. (a) Discuss in detail about problems in code generation. (10)
   (b) Explain in detail about the peephole optimization. (10)

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DE–2808 **305**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

MULTIMEDIA SYSTEMS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) What are the five basic components of the early multimedia system? Explain. (10)

(b) What are the different types of development tools? Explain. (10)

2. (a) Explain the four broad types of hypermedia system. (10)

(b) What do you mean by cross-platform compatibility? Explain the three ways of implementing cross-platform compatibility. (10)

3. (a) Describe the factors influencing the choice of a suitable storage system for multimedia. (10)

(b) Write a brief note on the following : (10)

(i) RAID

(ii) Optical media.

4. (a) Discuss briefly about compact disk-digital audio (CD-DA) and compact disk-read-only memory (CD-ROM). (10)

(b) List and explain any two standards used for compressing bitonal images. (10)

5. (a) Draw the block diagram of a sound system and explain the working principle. (10)

(b) Discuss briefly about musical instrument digital interface (MIDI). (10)

6. (a) What are the three different coding schemes recommended by the MPEG committee? Explain. (10)

(b) What are the two formats adopted by ITU-T for video telephony? Explain. (10)

7. (a) Describe the four stages used in developing a multimedia application. (10)

(b) What are the four important features of object-oriented system? Explain. (10)

8. (a) Describe the four main applications of interactive TV. (10)

(b) Discuss briefly about the following : (10)

(i) Integrated Services Digital Network (ISDN)

(ii) Asynchronous Transfer Mode (ATM).

DE–3207 **106**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

LAB I — DATA STRUCTURES USING 'C'

(2001 and 2002 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Examiner has to choose the question for each candidate by lot system.

Each subdivision carries 50 marks.

Break up of marks :

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. (a) Write a C program to read the order and elements of matrix A and matrix B and find their addition to get matrix C. Display all the matrices A, B and C.

   (b) Write a C program to implement push and pop operations on stack.

Cut here

2. (a) Write a C program to implement all string operations using switch statement.

   (b) Write a C program to sort 10 numbers in ascending order.

3. (a) Write a C program to find the factorial of a given number using recursion.

   (b) Write a menu driven program in C to find an element using linear and binary search methods.

Cut here

4. (a) Write a C program to copy contents of one file to another file.

   (b) Write a C program to read two matrices and find matrix subtraction.

Cut here

5. (a) Write a C program to read 10 values in an array variable. Use pointers to locate and display each value.

   (b) Write a C program for linked list implementation of queue operations.

Cut here

6. (a) Write a C program to read 10 numbers and store them in an array. Find the largest and smallest numbers after rearranging them.

   (b) Write a C program to implement string concatenation operation using two strings.

7. (a) Write a C program to evaluate the given mathematical expression using stack.

   (b) Write a C program to read a students name, reg. number and three subjects marks. Find total and average. Also find grade using the following condition.

Average	Grade
> 80	distinction
< 80 but > 60	I class
< 60 but > 40	II class
Else	fail

Display the result in a neat format.

DE–3208 **107**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

LAB II — MS-OFFICE

(2001 and 2002 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Candidate has to select ONE question by lot system.

1. (a) Type the Document regarding Information Technology and do the following.
   - (i) Find and Replace the word using wild card and case matching.
   - (ii) Extract some paragraph to another File.
   - (iii) Do spell check in the document and add new word in the Dictionary.
   - (iv) Change the font size and underline the paragraph.
   - (v) Use Bullet and Numbering.

(b) Prepare three slides for showing types of computers with the following settings.
   - (i) Set Different Slide Transition.
   - (ii) Give Header and Footer for Each Slide.
   - (iii) Set Slide Timing for Each Slide.

2. (a) An Invoice table has the following Details.

   Name of the Item, Price of each Item, Number of Quantity, Item Purchase Discount rate.
   - (i) Type the Data in the above format.
   - (ii) Find the Total Price, Average Price using functions.
   - (iii) Sort the name in alphabetical order.
   - (iv) Draw a pie chart for total price.

(b) Prepare three slides in power point showing the features of computer and also set timings to view it.

3. (a) Type the four c Cut here e row values in a document and convert into table format with headings.

(b) Create a work sheet with the following Data for Five Students.

   Regno, Name, Four Test Marks.
   - (i) Find the Total, Average of each student.
   - (ii) Find the highest and lowest scores in each subject.
   - (iii) Draw the bar chart for the Average Mark.
   - (iv) Declare the Result is “PASS" if average is more than or equal to 50% otherwise “FAIL".

4. (a) Prepare three slides for showing the features of MS-office with the following settings.
   - (i) Set Different Slide Transition.
   - (ii) Give Header and Footer for Each Slide.
   - (iii) Set Slide Timings for Each Slide.

(b) Construct the worksheet with the following fields.

   Empno, Name, Designation, HRA, DA, Basic-pay, Gross-pay, Net-pay.
   - (i) Type Five Data for Empno, Name, Designation, Basic-pay.
   - (ii) Calculate :

     DA = 35% of Basic-pay.

     HR = 15% of Basic-pay.

     Gross-Pay = Basic-Pay + DA + HRA.

     Net-Pay = Gross-Pay -(HRA).

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5. (a) Create your class time table for five working days using table in MS-Word. Split the Days coloumn and sort it. Demonstrate Insertion and Deletion of row coloumns. Invoke the table in to MS-Excel.

(b) Prepare three slides with a text and picture expressing the introduction of new product.

6. (a) Prepare three slides in PowerPoint showing the features of MS-Office and also set timings to view it.

(b) Create student table with the following fields in MS-Access.

Field Name	Type
REGNO	Number
NAME	Text
SEX	Text
AGE	Number
DEGREE	Text
PERCENTAGE	Number.

(i) Add Five Records using Forms.

(ii) Display the records for SEX = “M" AND Percentage > 60.

(iii) Display the records for SEX = “F" AND Percentage < 50.

(iv) Sort the table in the Descending order of percentage.

DE–3209 **113**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

LAB III – GRAPHICS USING C++

(2001 & 2002 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Each subdivision carries 50 marks.

Candidate has to select one question using lot system.

1. (a) Write a C++ program to show graphical transformation namely transformation of two dimensional object.

   (b) Draw any 3-D object and perform animation on it.

Cut here

2. (a) Write a C++ pr[illegible] non–liner objects arcs and curve.

   (b) Draw a ball. Perform animation on it.

3. (a) Draw a circle. Divide into four quadratic and fillup different colour for each quadrant.

   (b) Write procedure and a C++ program to draw non–linear object circle and three concentric circle.

Cut here

4. (a) Write a C++ [illegible] anipulate the graphical object using SHADING technique.

   (b) Write a Procedure and a C++ program to draw nonlinear object ellipse.

Cut here

5. (a) Write a C++ [illegible] w a straight line using Simple Brehnhams line algorithm.

   (b) Draw a cube and specify its co–ordinates.

Cut here

6. (a) Write a C++ program to draw a straight line using simple DDA techniques.

   (b) Write a C++ program to show graphical transformation on three dimensional objects.

Cut here

7. (a) Write a C++ p[illegible] linear objects lines, rectangle using Switch case option. First drawn the procedure and then implement the task.

   (b) Write a C++ program to show graphical transformation rotation of a two dimensional object.

8. (a) Write a C++ program to show graphical transformation scaling of a two dimensional object.

   (b) Write a C++ program to draw a straight line using simple DDA technique.

DE–3210 **114**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

LAB IV — ALGORITHMS

(2001& 2002 batch)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Each subdivision carries 50 marks candidates has to select ONE question using lot system.

Break up of marks

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

1. (a) Write a C program for insertion sorting.

(b) Write a C program for 8–puzzle problem using backtracking.

2. (a) Write a C prog selection sorting.

(b) Write a C program to solve traveling salesman problem using branch and bound method.

3. (a) Write a C prog ort (Divide and conquer)

(b) Write a C program for Binary searching.

4. (a) Write a C program for straight exchange sorting.

(b) Write a C program for Recursion (Factorials and Fibonacci)

5. (a) Write a C prog rt.

(b) Write a C program fo ng.

6. (a) Develop a C pr problem using working backward method.

(b) Write a C program to implement the basic operations in stack.

7. (a) Write a C prog various operations in dequeue.

(b) Write a C program to solve traveling salesman problem using Heuristics method.

8. (a) Write a C prog four problem using backtracking method.

(b) Write a C program to solve towers of Honor.

DE–3211 **106**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

LAB I — DATA STRUCTURES USING C AND MS OFFICE

(2003 and 2004 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Examiner has to select TWO questions choosing ONE from C programs and another from MS-Office.

C-PROGRAMS

1. (a) Write a C program to sort a set of elements using Insertion sort.

(b) Write a C program to copy contents of one file to another file.

2. Develop a 'C' program [illegible] 'push' and 'pop' operations on stack.

3. Write a C program [illegible] insert and delete operations on Linked List structures.

4. Write a C program to [illegible] of a quadratic equation $ax^2 + bx + c = 0$.

5. Write a C program to [illegible] string operations using 'Switch' statement.

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[illegible]CE PROGRAMS

6. Using MS-Word, prepare your own biodata with the help of the template and use numbering and bullets wherever necessary.

7. Prepare three slides [illegible] with a text and picture expressing the introduction of new product.

8. Create worksheet in MS-EXCEL with the following details :

Name, description of the item, quantity purchased, stock in hand, re-order level.

(a) Enter the 5 data in the above format.

(b) Calculate amount = price * quantity purchased.

9. Using MS-Word, type the 4 columns and five rows values in a document and convert into table format with [illegible]

10. Prepare five slides in PowerPoint with a text and picture expressing the courses offered in your Univ[illegible] animation effects.

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DE–3212 **113**

DISTANCE EDUCATION

M.C.A.(NS) DEGREE EXAMINATION, DECEMBER 2011.

GRAPHICS USING C++ AND ALGORITHMS

(2003 and 2004 Batches)

Time : Three hours Maximum : 100 marks

TWO questions are to be selected by examiner and the candidate should select the question by lot system.

1. Write a C++ program to show graphical transformation rotation of a two dimensional object.

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2. Draw a circle Divide i tic and fill up different color for each quadrant.

Cut here

3. Draw any 3-D object a imation on it.

Cut here

4. Write a program to sc Salesman problem using Heuristics Method.

Cut here

5. Write a program for I g.

Cut here

6. Write a program to sc onoi.

7. Write a program for converting a decimal number into binary system. Use conversion to octal system as a SUBGOAL.

Cut here

8. Write a program for A tion using Recursion.

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DE–3213 **206**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

LAB V — RDBMS

(2001 & 2002 Batches)

Time : Three hours Maximum : 100 marks

Break up of marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Each question carries 100 marks.

Examiner has to choose and give ONE question for each candidate by lot system.

1. Write a PL/SQL program to
   (a) Enter 10 students information such as rollno, marks in 5 subjects.
   (b) Find total marks, average marks and result for each student and count the number of passes per subject and display the percentage of passes per subject.
   (c) Manipulate records such as adding, deleting and counting number of records.

2. Write a PL/SQL program Cut here anagement system.
   It should give a report which includes customer details, A/C details, Deposit, with drawals, cheque collections, branch location, city etc.
3. Write a PL/SQL program for to create a mailing table which contains the address in the following format :
   NAME :
   STREET :
   PLACE :
   DISTRICT :
   PINCODE :
   Given a district name, print all the mailing address in that district.
   Also given a name of person, delete the particular mailing address from the table.

Cut here

4. Create a form for seat re roblem. Check validity of each field.
   Use procedures and functions.
   General reports for
   (a) Reservation details
   (b) Cancellation details.

Cut here

5. A hospital maintains blo ecord information in a file, The items are
   Donor number, Donor name, Donor Age, Donor Address, Pin, Place of birth, blood group (A, B, B^+, O^-)
   Write a PL/SQL program to print the numbers, name of the donor for the below categories :
   (a) blood donor having the blood group B^+
   (b) blood donor in the age group between 25 to 30
   (c) male donors having blood group B in the age between 25 to 30.
6. Write a program in SQL to departmental billing system, having the following fields.
   Product -id, Product name, Price, Qty, Customer name, Customer Address, Phone no, Credit/Cash.

Write a report program for the following conditions :

(a) The customers, who are choosing credit billing.

(b) The customers, who are choosing cash billing.

Cut here

7. Write a PL/SQL program to calculate income tax of employees based on the below conditions.
   - Gross pay between 1,50,000 to 2,50,000
     IT is 15% of the pay
   - Gross pay between 2,50,001 to 5,00,000
     IT is 1000 + 20% of the pay, above 2,50,000
   - Gross pay above 5,00,001
     IT is 3000 + 30% of the pay which is above 5,00,000.

   Prepare a report that contains the employee details, basic pay, allowances, deductive, gross pay, net pay and income tax.

Cut here

8. Write a PL/SQL for inventory management system. It should contain the following details.
   - Product details
   - Product id
   - No. of products in the stack
   - Supplier information
   - Re-order level
   - Product purchase.

DE–3214 **207**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

LAB VI — SHELL PROGRAMMING

(2001 and 2002 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Break-up of marks :

Record Note Book : 5
Algorithm : 20
Program : 30
Debugging and Execution : 35
Result : 10
Total : 100

Examiner to select and give ONE question to each candidate.

Each subdivision carries 50 marks.

1. (a) Write a shell program to convert all capital letters in a file to small letters.

(b) Display the list of last [illegible] in the current directory. Also store this list in a file profile.

Cut here

2. (a) Merge the contents of the files file1, file2 and file3. Sort them and display the sorted output on the [illegible] page.

Cut here

(b) Write a shell program to sum up the series :

$$\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\cdots+\frac{1}{10!}$$

3. (a) Write a shell program for a file with each record containing name of city, name of state and name of country. Sort this file with country as the primary sort key and state as the secondary sort key.

(b) Write a shell program to display the result 'PASS' or 'FAIL' using the information given below :

Student Name, Student Reg. No, Mark1, Mark2, Mark3, Mark4. The minimum for passing in each subject is 50.

4. (a) Suppose a user [illegible] some files in current directory using a command like MV filename filename.$$. Write a shell script to search all such files and rename them such that they do not contain the shell PID.

Cut here

(b) Write the shell script which gets executed the moment the user logs in. It should display the message "GOOD MORNING" / "GOOD AFTERNOON" / "GOOD EVENING" depending upon the time at which the user logs in.

5. (a) Write a shell script which will receive either the file or the file name with its full path during execution. This script should obtain information about this file as given by 1s-1 and displays it in proper format.

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(b) Write a shell program using 3 arguments to take the pattern as well as input and output file names. If the pattern is found, display "pattern found", else display "Error message". Also check if right number of arguments are entered.

6. (a) Write a menu driven shell program to copy, edit, rename and delete a file.

(b) Write a menu driven shell program to perform the following tasks :

(i) Enter a sentence in file.

(ii) Search a given whole word in an existing file.

(iii) Quit.

7. (a) Merge the con[illegible] of the files file1, file2 and file3. Sort them and display the sorted output on the screen, page by page.

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(b) Display the list of last 20 files present in the current directory. Also store this in a file profile.

8. (a) Write a shell p[illegible]

Cut here

$$\frac{1}{1!}+\frac{1}{3!}+\frac{1}{5!}+\cdots+\frac{1}{n!}.$$

(b) Write a menu driven shell program to copy, edit, rename and delete file.

DE–3215 **213**

DISTANCE EDUCATION

MASTER OF COMPUTER APPLICATIONS (NS) DEGREE EXAMINATION, DECEMBER 2011.

Lab VII — INTERNET PROGRAMMING

(2001 and 2002 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

Examiner has to choose ONE question and give it to each candidate by lot system

1. (a) Write a java script to create an order form to select the house articles.

   (b) By using database connectivity display the records in a table.

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2. (a) Write a java program using applet to display the dialogue and menu in applet.

   (b) Develop a picture gallary having atleast 3 pages, each of them is having several pictures.

3. (a) Write a VB script to do the following.

   (i) Display the current date.

   (ii) Find the difference between the two dates.

   (iii) Find the age of a person by providing date of birth.

   (b) Write a swing program to create the tabbed pannels.

4. (a) By using database connectivity insert and delete a records from a table.

   (b) Using atleast 2 [illegible] create a screen with a string "Web Design".

Cut here

5. (a) Write a java program using applet to create frames and its controls.

   (b) Create a web page in the format of front page of a news paper using text links. Align the text with colors.

Cut here

6. (a) Write a swing program to create buttons with

   (i) Tool tip text

   (ii) Image

   (iii) Border

   (iv) Short cut key

   (b) Using atleast 15 HTML tags, create a screen with a string "ALAGAPPA UNIVERSITY".

7. (a) Design and publish web page for travel agency.

   (b) Write a java script to create a window by using the confirm message.

8. (a) Design and publish a web page for a college.

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(b) Write a java program to set a color as the back ground.

DE–3216 **214**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

LAB VIII — VISUAL PROGRAMMING

(2001 and 2002 batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Break up of marks :

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Each candidate should select ONE question using lot system.

1. (a) Write an event procedure for the following
   - (i) Display date and time in the label box at run time
   - (ii) Find the power of a number
   - (iii) Temperature conversion
   - (iv) Determine whether the given string is palindrome or not.

   (b) Write a VC++ program to create a window of desired size using MFC.

2. (a) Create a form with textbox, combobox and command button and do the following operations.
   - (i) Add the university name in the combobox at run time
   - (ii) Search and delete the particular university in the combobox.

   (b) Write a VC++ program to handle windows messages in MFC program.

3. (a) Create a VB pr Cut here ntrol array, to perform the following operations on a simple calculator.
   - (i) Addition
   - (ii) Subtraction
   - (iii) Multiplication
   - (iv) Division
   - (v) Square
   - (vi) Square root
   - (vii) Modulus
   - (viii) Power
   - (ix) Percentage

   (b) Write a VC++ program to fill background of the client area with a bit map.

4. (a) Create a VB pr Cut here [SFlex grid control, display the multiplication and addition table of 20 rows and columns.

   (b) Write a VC++ program to get the status of the shift and toggle keys using MFC.

5. (a) Write a VB program using built in AX control develop the windows notepad with file and edit menu operation and also display the floating menu whenever necessary.

   (b) Write a VC++ program to generate a status bar and show the status of caps lock, num lock and scroll lock in it.

Cut here

6. (a) Create a VB pr ployee database and calculate the following :

HRA = 10% of BP
DA = 5% of BP
PF = 3% of BP
LIC = 5% of BP
GP = BP + DA + HRA
NP = GP – (PF + LIC).

Using remote data access object, develop the following operation

(i) Insert a record
(ii) Search and delete a record
(iii) Modify the record.

(b) Write a VC++ program to create a list box in a window.

Cut here

7. (a) Write a VB pr[illegible]rm the following operation in a record of random access file

(i) Insert
(ii) Delete
(iii) Search.

(b) Write a VC++ program to find out whether a mouse is attached or not; and if attached how many buttons are present in it.

8. (a) Create a VB program using AX DLLOR EXE add a class module that would perform the following function

(i) Test whether the given number is perfect or not
(ii) Find the factorial of the given number

(b) Write a VC++ program to create a window of desired size using MFC.

DE–3217 **206**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

LAB III — RDBMS AND SHELL PROGRAMMING

(2003 and 2004 Batches)

Time : Three hours Maximum : 100 marks

Examiner has to choose and give ONE questions to each candidate by lot system.

Break up of marks :

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. (a) A company ABC Pvt. Ltd. states monthly salary to its employee. It consists of Basic Pay (BP), allowances, Deduction.

DA – 43 of BP, HRA – 7% of BP + DA,

Deductions : PF - Subscribed by an capital, LIC - payable by an employee, the salary saving scheme and Loan recovery - if any payable by the employee Create a main table with 10 records which named as master, which contains. Employee number, name, designation, Basic Salary, HRA, DA, Bank A/c number, LIC Premium number.

A transaction file contains :

Employee number, PF subscription, LIC premium amount and Loan Recovery.

Write a program in SQL to prepare a report with the following information: Serial number, Employee number, Bank A/C number, Name, BP, Total Allowance, Gross Pay, Total deductions, Net pay.

(b) Write a Shell program to sum up the series

$$\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \ldots\ldots\ldots + \frac{1}{10!}.$$

2. (a) A hospital maintains blood donor's record information in a file. The items are Donor number, Donor name, Donor age, Donor address, Pin Code, Place of birth, blood group.

Write a program to print out the number, name and address of the donors for the following categories :

(i) Blood donor having the blood group AB

(ii) Blood donors in the age group between 16 to 25.

(iii) Female donors having blood group A in the age between 20 and 30.

(b) Write a shell program to convert all capital letters in a file to smallcase letters.

Cut here

3. (a) Each morning r vendor buy newspapers in wholesale from a distributor for 60 paise. He sell them in retail for 75 paise. At the end of the day the unsold papers are returned to the distribution for a 30 paise rebate per paper. Write Oracle program to prepare a report for the newspaper vendor in the following format with 10 weeks data.

Week	Bought	Sold	Return	Profit	Loss
1	50	30	XX	XX	XX
2	75	62	XX	XX	XX
....	....	....	....	....	....
....	....	....	....	....	....
Total					

(b) Write a shell program for a file contains records with each record containing name of city, name of state and name of country. How would you sort this file with country as the primary sort key and state as the secondary sort key.

4. (a) A company ABC Pvt. Ltd. states monthly salary to its employee. It consists of Basic Pay (BP), Allowances, Deduction.

DA-43 of BP, HRA – 7% of BP + DA,

Deductions : PF – Subscribed by an capital, LIC – payable by an employee, the salary saving scheme and Loan recovery – if any payable by the employee Create a main table with 10 records which named as master, which contains. Employee number, name, designation, basic Salary, HRA, DA, Bank A/c number, LIC Premium number.

A transaction file contains :

Employee number, PF subscription, LIC premium amount and Loan Recovery.

Write a program in SQL to prepare a report with the following information : Serial number, Employee number, Bank A/c number, Name, BP, Total Allowance, Gross Pay, Total deductions, Net pay.

(b) Write the shell script which gets executed the moment the user logs in. It should display the message "Good morning"/ "Good afternoon"/ "Good Evening" depending upon the time at which the user in.

5. (a) The hospital billing system consists of the following fields :

Patient number, Patient name, Age, Doctor attending, patient–type (In/Out), Consulting charge, Blood test charge, X-ray charge, other test charges and total fee.

Write a menu driven program in SQL for the following conditions :

(i) List of the patients, undergone blood test and charges.

(ii) List of the patients, who have taken X-rays with charges.

(iii) List of the patients, who belong to In-patient category.

(iv) List of the patients with total fee.

(v) Exit.

(b) Suppose a user has renamed some files in current directory using a command like my filename filename $$. Write a shell script to search all such files and rename them such that they do not contain the shell PID.

6. (a) An examination has been conducted to a class of 7 students and four scores of each student have been provided in the data along with register number, name.

Write SQL program to do the following :

Assign a letter grade to each student based on the average score and list out the students register number and average score, grade. The minimum pass for each student is 50.

The grading system :

Average Score	Grade
90-100	A
75-89	B
60-74	C
50-59	D
Less than 49	F-Fail

(b) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it :

(i) File existence

(ii) File readable

(iii) File writeable

(iv) Both readable and writeable.

7. (a) Write a program in SQL to hospital billing system having the following fields Patient number, Patient name, Age, Doctor attending, patient-type (In/Out), Consulting charge, blood test charge, X-ray charge, other test charges and total fee.

Write a report program for the following conditions :

(i) The patients, undergone blood test.

(ii) The patients, who have taken X-rays.

(iii) The patients, who belong to In-patient category.

The common fields to be included in the above mentioned reports are, patient number, patient name, Age and corresponding charge and total fee.

(b) Write a shell program to display the result PASS or FAIL using the information given below :

Student Name, Student Reg. No, Mark 1, Mark 2, Mark 3, Mark 4.

The minimum pass for each subject is 50.

8. (a) An airline reservation database contains the reservation table and personal table

Reservation Table : Flight number, Passenger Name, Seat Number.

Personal Table : Passenger Name, Sex, Age,
Maritial Status, nationality.

Write a program in ORACLE to prepare the following reports.

(i) List the passengers names with seat number according to flight number wise.

(ii) Total number of married female candidates in a particular flight.

(iii) List out the female candidates in between the ages of 18 to 22 for all flights.

(b) Suppose a user has renamed some files in current directory using a command like my filename filename $$. Write a shell script to search all such files and rename them such that they do not contain the shell PID.

DE–3218 **213**

DISTANCE EDUCATION

M.C.A. (N.S) DEGREE EXAMINATION, DECEMBER 2011.

LAB IV – INTERNET AND VISUAL PROGRAMMING

(2003 & 2004 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Examiner has to choose the ONE question for each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a JavaScript to create a order Form to select the house articles.
   (b) Write a Visual C++ program to create a List Box in a Window.

Cut here

2. (a) Write a VB scr owing
      (i) Display the current date
      (ii) Find the age of a person by providing date of birth
      (iii) Find the difference between the two dates.
   (b) Write Visual C++ program to fill background of the client area with a bitmap.

Cut here

3. (a) By using datab (JDBC) display the records in a table.
   (b) Write a Visual C++ program to create a window of desired size using MFC.
4. (a) Write a Java program using Applet to display the dialogue and menu in applet.
   (b) Write a visual C++ program to handle windows messages in MFC program.

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5. (a) Write a Java p Applet to create the frames and its controls.
   (b) Write a visual C++ program to get the status of the shift and toggle keys using MFC.

Cut here

6. (a) Develop a singl ement for a shop to be opened shortly.
   (b) Write a visual C++ program to generate a status bar and show the status of Caps Lock, NUM Lock and Scroll Lock in it.

Cut here

7. (a) Design and publish a web page for a college.
   (b) Write a Java program using Applet to display the different colors and fonts.

Cut here

8. (a) Develop a Web ruitment agency in an IT industry.
   (b) Write a swing program to create the Tabbed Panels.

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DE–3220 **307**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

LAB X — COMPILER DESIGN

(2001 and 2002 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Break up marks :

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Candidate has to choose ONE question using lot system.

Each question carries 100 marks.

Cut here

1. Write a C program which accepts $(1/0)^{+}.10(1/0)^{*}$ draw the transition diagram for DFA.

Cut here

2. Construct a C program [illegible] string and check whether it satisfies the following production rule :

(By using operator precedence Grammar)

S → AbB

A → aBb

B → a ¦ Bba

Write a C program to recognise identifier in the given source code.

Cut here

3. Write a C program to rec[illegible] in the given source code.

Cut here

4. Write a C program to [illegible]ng and check whether it satisfies the following production rule. (By using shift reduce parsering).

E → E + E ¦ E – E ¦ E ∗ E ¦ E/E ¦ id ¦ Const ¦ (E)

Cut here

5. Write a C program to tra[illegible]metic expression into postfix notation.

Cut here

6. Write a C program t[illegible]e expression $-(a+b)*(c+d)+(a+b+c)$ into quadruples and triples.

Cut here

7. Write a C program to con[illegible]sing table for the following grammer.

S → Aa ¦ bAc ¦ Bc ¦ bBa

A → d

B → d.

DE–3221 **306**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, DECEMBER 2011.

LAB V — MULTIMEDIA SYSTEMS AND COMPILER DESIGN

(2003 and 2004 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Answer both subdivision (a) and (b).

Examiner has to choose the question for each candidate by lot system.

Each subdivision carries 50 marks.

Break up of marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

1 (a) Design an advertisement for an educational institution using flash.

(b) Write a C program which accepts the regular expression (a/b)(a/b) draw the transition diagram for DFA.

2. (a) Create a flash movi movement using Motion tweening.

Cut here

(b) Write a C program to construct LMD and RMD for the following Grammar.

S - > a/S/ (R)

R - > R, S/S

(i) (c, (c, c))

(ii) To find the LEADING AND TRAILING sets.

3 (a) Create a flash movie to display the masking effects.

(b) Write a C program to construct an LALR parsing table for the following grammar.

S->XX

X->yX/x .

4 (a) Create a flas display cell animation.

Cut here

(b) Write a C program to convert Infix to postfix notation.

5 (a) Create a Pho ge, to display Filter effects.

Cut here

(b) Write a C program to check the production rule whether it is Operator grammar or not.

(i) S->aTbd, S->aTdc

(ii) E->E+E/ E/E /E /id

6 (a) Create a Pho ge, which displays the lighting effects.

Cut here

(b) Write a C program to implement the Semantic and Syntatic Errors.

7 (a) Create a Fla for slide presentation.

Cut here

(b) Write a C program to construct the PARSE Tree for the following expression (3 * 2 + 2)*4.

8 (a) Create a flas at performs the Shape tweening.

Cut here

(b) Write a C program to translate the expression —(a+b)*(m+n)(m+n+a) into postfix notation and three-address code.